



Examiners' Report Principal Examiner Feedback

January 2025

Pearson BTEC Nationals in
Applied Science/Forensic and Criminal
Investigation (31617H)
Unit 1: Principles and Applications of
Science I

Edexcel and BTEC Qualifications

Edexcel and BTEC qualifications come from Pearson, the world's leading learning company. We provide a wide range of qualifications including academic, vocational, occupational and specific programmes for employers. For further information visit our qualifications website at: <http://qualifications.pearson.com/en/home.html> for our BTEC qualifications.

Alternatively, you can get in touch with us using the details on our contact us page at <http://qualifications.pearson.com/en/contact-us.html>.

If you have any subject specific questions about this specification that require the help of a subject specialist, you can speak directly to the subject team at Pearson. Their contact details can be found on this link: <http://qualifications.pearson.com/en/support/support-for-you/teachers.html>

You can also use our online Ask the Expert service at: <https://qualifications.pearson.com/en/contact-us.html>

You will need an Edexcel Online username and password to access this service.

Pearson: helping people progress, everywhere

Our aim is to help everyone progress in their lives through education. We believe in every kind of learning, for all kinds of people, wherever they are in the world. We've been involved in education for over 150 years, and by working across 70 countries, in 100 languages, we have built an international reputation for our commitment to high standards and raising achievement through innovation in education. Find out more about how we can help you and your learners at: <http://www.pearson.com/uk>.

Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this [link](#).

Introduction

Biology

This examination allows learners to show their knowledge and understanding about the structure and functions of cells and tissues. Centres are ensuring that their learners are aware of the content being tested using the additional guidance document and practicing questions using the past papers available. This approach with learners has meant that fewer blank spaces were seen, and many more learners were achieving marks across the whole of the paper.

Chemistry

This paper is one of three externally assessed papers, worth 30 marks each, that assess Unit 1 – Principles and Applications of Science 1. This paper covers key chemistry concepts including atomic structure and bonding and chemical and physical properties of substances to their uses.

Physics

The examination was designed to allow learners to show their knowledge and understanding of 'Waves in Communication'. The topics included fundamentals of wave behaviour, understanding graphical representation of waves and emission spectra. The applications included communication, endoscopes and digital versus analogue signals.

Overall Performance of the Unit

Biology

Learners that were successful this year were able to recall the magnification equation and carry out the calculation using clear and logical working.

It is beneficial for learners to appreciate the requirement of each of the command verbs so that they can target their response and provide appropriate credit worthy answers. Explain is the command verb which seems least understood, especially for the 2-mark question about the speed of conduction. Learners tend to make an identification point without a linked expansion for full marks,

Learners should ensure that they have read the questions carefully so that they can give an appropriate answer, depending on if the focus of the question was structure or function, or the relationship between the two. Learners still struggle to recall key facts, such as the structure of a plasmid, and they seem to find it difficult to provide links between different related aspects of the specification, such as respiration and the function of ciliated epithelial cells.

Most learners were able to access the extended open response question and provide at least one suggestion of how a drug could be used to treat depression. Some learners had sufficient relevant knowledge and understanding to discuss several different mechanisms clearly, logically and coherently.

Chemistry

Learners that were successful this year showed a good understanding of key terms, scientific vocabulary and mathematical skills and were able to apply this knowledge and understanding to new scenarios. They engaged with the question and used the prompts in the questions to inform their answers. They were precise with their language, were able to link ideas and explain in detail where necessary. Learners that performed well, were able balance equations, draw dot and cross diagrams for ionic compounds and carry out calculations in a clear and logical manner which helped them to arrive at the correct answer.

Physics

Learners were generally able to recall that waves of the electromagnetic spectrum all travel at the same speed and could correctly annotate diagrams to show the amplitude of a wave. Most were also able to correctly illustrate the path of a light ray within an optical fibre.

Learners were able to demonstrate their ability in completing calculations, although some learners still had difficulty in correctly rearranging equations, with many also struggling in being able to correctly convert units and perform more complex mathematical operations. Knowledge of the purpose of cladding in an optical fibre was not clearly understood, and many learners struggled to explain the significance of the lines in an emission spectrum.

Although learners were familiar with the use of Bluetooth® technology in communication and could suggest a limitation of its use, many struggled to explain that limitation using knowledge of wave properties and behaviour.

Many learners were able to describe graphical trends shown for analogue and digital signals, but few were able to relate the trends seen to their knowledge of these types of signals and explain the differences shown for each type of signal.

Individual Questions

Biology

Question 1

1(a) – This question was a multiple-choice question which required the learners to identify the stage of the action potential from the graph in Figure 1. Most learners were able to correctly identify that the stage was depolarisation.

(a) Identify the stage of the action potential labelled X in Figure 1.
(1 mark)

- ☒ A depolarisation
- ☐ B hyperpolarisation
- ☐ C refractory period
- ☐ D repolarisation

1(b) – Many learners correctly gave the ion sodium. There were some common incorrect answers such as 'calcium' and 'potassium'. 1 mark was awarded to this response for sodium. The learner could also give the periodic table symbol for sodium, Na⁺, with or without the charge.

(b) State which ion enters the neurone during stage X of Figure 1.

(1)

Sodium

1(c) – This question was a multiple-choice question which required the learners to recall the threshold membrane potential.

(c) Identify the threshold membrane potential that must be reached to generate an action potential.
(1 mark)

- ☐ A -70mV
- ☒ B -55mV
- ☐ C 0mV
- ☐ D 30mV

1(d) – This question asked the learners to explain the effect of myelination of the speed of conduction. The learners should have stated the effect as the identification point for this explanation and then given a reason for the effect to fulfil the demand of the command verb.

Most learners could state that the impulse was faster. Although there were some learners with a misconception regarding the myelin sheath acting as an obstacle and therefore reducing the speed of conduction.

A common mistake was for learners to repeat the stem of the question and explain that the speed increases because the axon is myelinated/surrounded by myelin sheath.

A high number of learners suggested that the increased speed of conduction was due to better insulation rather than saltatory conduction. Some learners suggested that it was the neurone itself that jumped rather than the signal.

The best responses simply stated 'faster due to saltatory conduction' or gave a description of jumping between the gaps in the myelin.

This response was awarded 2 marks. The idea of increased speed or faster was acceptable for MP1 (however, just fast would not be enough, there needs to be a comparison such as faster, quicker, increased). The second marking point can also be awarded. Jumping along the neurone was not enough without the reference to the nodes of Ranvier, or reference to gaps in the myelin (which was sufficient as a description of the nodes of Ranvier).

An action potential travels through an axon.

Axons can be myelinated or unmyelinated.

(d) Explain the effect of myelination on the speed of an action potential.

(2)

myelinated neurons have a faster action potential as the signal jumps along the neuron across the nodes of Ranvier. This gives it a higher action potential.

The next response was awarded 1 mark. The learner has given the identification point but has not provided the correct linked expansion for the explanation.

(d) Explain the effect of myelination on the speed of an action potential.

(2)

The speed of an ~~the~~ action potential will be faster. This is because myelinated axons are insulated which means the speed of an action potential will be faster than a unmyelinated axon.

Question 2

2(a) – Many learners gained all 3 marks for the calculation of the magnification of the observed cell. The most common error was an incorrect conversion, which gave a power of ten error to the final answer.

Some learners incorrectly recalled the magnification equation or incorrectly used the magnification triangle and so inverted the division and gave a magnification of x4. Learners should be encouraged to show their working and then sense check their final evaluation for the calculation.

This response was awarded 3 marks for x250.

(a) Calculate the magnification used to observe the cell.

(3)

Show your working.

$$\text{magnification} = \frac{\text{Image Size}}{\text{actual size}}$$

$$3.2 \text{ mm} \times 1000 = 3200 \mu$$

$$\text{magnification} = \frac{3200}{12.8} = 250$$

$$\text{magnification} = \times \underline{250}$$

The next response was awarded 2 marks. The learner has not attempted the conversion so the two values were not in the same unit, but they have correctly substituted and evaluated the magnification equation and therefore 2 marks were awarded as error carried forward marks.

(a) Calculate the magnification used to observe the cell.

(3)

Show your working.

$$\frac{3.2}{12.8} = 0.25$$

$$\text{magnification} = \times \underline{0.25}$$

2(b) – This question was challenging for the majority of learners as the question required the learners to make the connection between the function of the mitochondria and the cilia on ciliated epithelial cells.

Many learners were able to link mitochondria as the site of respiration or energy release or production. They were also able to link the need for energy to move/waft mucus therefore marking point 2 and marking point 3 were awarded.

Some learners gave the third marking point in several ways, such as “produce energy in the form of ATP from respiration”, but without acknowledgement of the high numbers of mitochondria from the question and therefore were not awarded the marking point for the appreciation that a lot of/high levels of ATP are required by the ciliated epithelial cells.

References to powerhouse of the cell commonly were seen. “Powerhouse of the cell” is a term that should not be used with learners – learners remember it, but it is not creditworthy phrasing.

A significant number of learners made no mention of ATP or incorrectly stated that the energy was needed for respiration.

There seemed to be a common misconception that the ciliated cells produced the mucus rather than move the mucus.

Some learners seemed to confuse the sperm cell with the ciliated epithelial cell and suggested that it is moved around the body.

This response was awarded 3 marks. The learners had correctly stated that the high numbers of mitochondria provide enough energy in the form of ATP for the movement of the cilia.

(b) Explain why ciliated epithelial cells contain high numbers of mitochondria. (3)

Ciliated epithelial cells contain a number of mitochondria in order to have enough energy to keep the constant swaying of cilia hairs. This will be because cilia hairs actively brush dust and unwanted pathogens from entering the lungs which requires continuous movement. This will mean high amount of ATP needs to be hydrolysed for the energy needed.

(Total for Question 2 = 6 marks)

The next response was awarded 1 mark. The learner had given the identification point for the explanation but not given the linked reasons for the high number of mitochondria.

(b) Explain why ciliated epithelial cells contain high numbers of mitochondria.

(3)

Ciliated epithelial cells contain high mitochondria
as they require a lot of energy.

Question 3

3(a) – Many learners correctly identified protein synthesis. The learner phrasing of making/creating proteins was also accepted.

There were a number of incorrect references to storing or transporting proteins and some learners suggested that the ribosomes were the site of respiration or photosynthesis.

This response was awarded 1 mark. The learner has correctly stated the function of the 70S ribosomes.

(a) State the function of the 70S ribosomes.

(1)

protein synthesis.

The next response was awarded 0 marks. This was a common incorrect suggestion of the function of the 70S ribosomes. A number of learners suggested that the ribosomes were involved in transport.

(a) State the function of the 70S ribosomes.

(1)

transport lipids + proteins

3(b) – Most learners were awarded at least 1 mark for this question, usually by stating that the plasmid is circular. Very few learners mentioned double stranded and so the second mark was usually for the DNA or genetic material component.

Some learners described the structure of the whole bacterium in the figure rather than the plasmid.

Learners should be encouraged to read the question carefully so that they are clear if the question is asking for the structure or the function – a number of learners gave detail about antibiotic resistance rather than the structure.

This response was awarded 2 marks. The learner had correctly described the structure of a plasmid.

(b) Describe the structure of a plasmid.

(2)

Circular double stranded DNA molecule

The next response was awarded 1 mark. The learner had not considered the shape or that the DNA is double stranded and so 1 mark was awarded for the DNA component. There wasn't a mark for the location of the plasmid within the cell, and so being loose or additional to the nucleoid was not creditworthy.

(b) Describe the structure of a plasmid.

(2)

Plasmids are the loose parts of DNA scattered around the prokaryote.

3(c) – Most learners were able to say that the number of Gram-positive bacteria decreased and therefore gained the first marking point.

It was not uncommon for learners to state that the number of Gram-negative bacteria increased due to the antibiotic treatment, rather than the correct idea that the antibiotic did not have an effect on the Gram-negative bacteria and therefore it could replicate and the number increased.

Some learners used key structural terminology correctly to be awarded marking point 3 and marking point 4. Learners were expected to explain that this antibiotic must be able to disrupt the peptidoglycan cell wall. Therefore, as the Gram-positive has a thick cell wall and no outer membrane for protection, this antibiotic was able to treat the Gram-positive bacterial infection.

There was the common misconception that the Gram-positive has a thin cell wall and therefore the antibiotic could enter the cell and destroy it.

Some learners described the Gram staining procedure or included the colours of stained bacteria in their response and so did not respond to the question. Learners should ensure that their answers are within the scope of the question, rather than answering with unrelated information about the topic.

This response was awarded 4 marks. The learner has correctly stated that the antibiotic treatment only works for Gram-positive bacteria, causing the number to decrease (described by the learner as diminish). The learner has then correctly explained this by referencing the disruption of the cell wall. The learner has also implied the idea of protection from the outer membrane in their final statement.

(c) Explain the effect of the antibiotic treatment.

Use Figure 4 to support your answer.

(4)
As seen in Figure 4, antibiotic treatment only works for gram positive bacteria as they are not antibiotic resistant like the gram negative bacteria. Over time the number of gram positive cells ~~decreases~~ begins to diminish as the antibiotic can easily disrupt the production of the peptidoglycan cell wall. However, the number of gram negative bacteria continues to grow as the antibiotic can't disrupt the production of the peptidoglycan cell wall due to it being sandwiched between a double membrane.
(Total for Question 3 = 7 marks)

The next response was awarded 2 marks. The learner has given marking point 1 about the decrease in Gram-positive bacteria and marking point 2, that the antibiotic has no effect on Gram-negative bacteria. However, the learner has not fulfilled the demand of the command verb by giving the linked explanation of these identification points.

(c) Explain the effect of the antibiotic treatment.

Use Figure 4 to support your answer.

(4)

The antibiotic was successful in stopping the synthesis of gram positive bacteria. This is seen in how the number of cells decrease after the antibiotic (penicillin) treatment.

However, the antibiotic had no effect on gram negative bacteria. This is seen in how in fig 4 the number of gram negative cells continues to steadily increase after antibiotic treatment and therefore needs phagocytosis.

(Total for Question 3 = 7 marks)

Question 4

4(a) – Overall this question was poorly answered by learners. Some learners were awarded 1 mark for stating that the cell wall is rigid or made of cellulose, but many incorrectly stated the cell could expand/was flexible or that the cell wall controlled the entry/exit of the water.

This response was awarded 2 marks. The learner had stated that the cell wall is made of cellulose, for marking point 1 and then they have provided a description of being able to withstand the pressure from the vacuole for the second marking point.

(a) Describe how the cell wall prevents the cell from bursting as the vacuole absorbs water.

(2)

The cell wall is made out of cellulose, this provides a rigid structure to maintain the shape of the cell. Therefore, the shape is able to be maintained even when the pressure of the vacuole is pushing against it.

4(b) – Overall this question was poorly answered by learners. Most responses only focused on a single idea, usually sunlight absorption, gas exchange or water absorption and failing to recognise that a 4-mark question would require more than one. The command verb, the number of marks and

the number of lines for the responses should be an indication to learners about the depth required in their response.

Where marks were scored it was almost always for maximising absorption of sunlight, connected to the moving of the chloroplasts to the edges of the cell.

Very few learners were able to identify that the movement of chloroplast would decrease diffusion pathway and increase CO₂ absorption.

Some learners incorrectly referred to the vacuole as a store of carbon dioxide, chlorophyll, or glucose.

Many learners incorrectly focused on the vacuole absorbing more water, increasing surface area and therefore increasing rate of photosynthesis.

A few learners included references to respiration in their answers, suggesting a confusion between the processes of respiration and photosynthesis.

This response was awarded 3 marks. The learner has credited the large surface area with being able to absorb more light energy- but "absorb more light energy" is a stand-alone mark and so marking point 2 can be awarded. The learner has also stated that more gases, eg. CO₂ can be diffused into the cell (marking point 4) and that there is a decreased diffusion pathway (marking point 3). Therefore 3 marks can be awarded overall.

(b) Explain how a large vacuole maximises the rate of photosynthesis.

(4)

The vacuole contains water and chemicals required for reactions. It absorbs carbon dioxide required to react with water in photosynthesis. It is very large, therefore increases the size and surface area of the cell. With a larger surface area, more light energy can be absorbed and more gases e.g. CO₂ can diffuse into the cell. It decreases the diffusion pathway.

The next response was awarded 2 marks. The learner has correctly linked the chloroplasts being pushed to the edge of the cell to the increased availability of sunlight. This is one identification point with a linked expansion and so 2 marks have been awarded.

(b) Explain how a large vacuole maximises the rate of photosynthesis.

(4)

A large vacuole maximises the rate of photosynthesis as it pushes the other subcellular organelles to the edge of the cell, including the chloroplasts. The chloroplasts is where photosynthesis occurs. Due to the chloroplasts being pushed to the edge of the cell, it means that they get more sunlight, especially as the palisade mesophyll is the top layer in leaves, which means the chloroplasts can occur have more sunlight available, so more photosynthesis ^

Question 5

This question was an extended open response 6-mark question, where learners should have explained different mechanisms of drugs. Credit was given for a range of different ideas which could increase the availability or action of serotonin or other neurotransmitters that are related to depression.

Most learners were able to state that a treatment for depression should increase serotonin levels, but few went on to discuss in detail how different drugs could be used to reduce depression.

The figure was provided as a reminder to learners about the synapse and the reuptake of the neurotransmitter and learners were able to use this to suggest ideas of increasing the concentration of serotonin in the vesicle, prevent reuptake, or suggesting that the drug would ensure that all serotonin was recycled by the brain.

Responses describing events at the synapse without discussion of how potential treatments may work were commonly seen. The most common treatment discussed involved increasing the amount of serotonin the synapse by giving this (or similar) as a drug directly to the patient. There was little appreciation for the requirement of a pre-cursor to cross the blood-brain barrier. There were a few mentions of agonist drugs, but these were

often confused with precursors and there was no explanation of how or why these were used. The term SSRI was used, but many learners could not explain how they worked.

Some responses discussed lifestyle modifications as treatments for depression, such as more exercise, better diet, improved sleep, and activities such as yoga, rather than discussing how drugs could be used to treat depression by acting on the synapse.

This response was awarded 6 marks. The learner has discussed several ways in which drugs could be used to treat depression.

Discuss different ways prescribed drugs could act at a synapse to treat depression.

Use the information in Figure 6 to support your answer.

(6)

Prozac is a drug that treats depression. The function of this drug is to block the reuptake pump in the presynaptic membrane. This stops the serotonin neurotransmitter being reuptaken ~~to~~ back into the ~~pre~~ axon terminal where it is ^{hydrolysed and recycled} ~~destroyed~~. Because the serotonin is not reuptaken, it ~~remains~~ remains in the synaptic cleft. This allows the serotonin to continually bind with the receptor sites on the postsynaptic membrane, increasing the number of transmissions of impulses. This helps treat depression ^{significantly} ~~well~~ because it removes the issue of a lack of serotonin in the synapse.

A different drug could be used which mimics serotonin. If it released mimics of serotonin into the synaptic cleft, it ~~not~~ could bind with the neurotransmitter receptors and trigger the transmission of a signal ~~without~~ ^{without} itself, ~~thereby~~ assisting serotonin and working with it to produce these transmissions.

This could also treat depression well as it allows more transmissions to occur without halting any existing functions of the ~~new~~ synapse.

Another drug could be used to stimulate the release of serotonin more.

This drug would stimulate the brain to release more serotonin hormone, removing the lack of serotonin in the synapse. This would mean there is ~~more~~ ^{more} ~~releasing~~ serotonin to send signals across a synapse, increasing the transmission of these signals thus treating the patient thus curing depression.

This response was awarded 3 marks. The learner discussed one way in which a drug could be used. One way discussed in correct detail was sufficient for a mark in level 2.

Discuss different ways prescribed drugs could act at a synapse to treat depression.

Use the information in Figure 6 to support your answer.

(6)

Some prescribed drugs may contain chemicals that increase the production of Serotonin in the brain. This means that if these tablets are regularly taken Serotonin levels ^{will} ~~may~~ increase in the ~~blood~~ body. Therefore the synapses in the brain will receive this ~~hormone~~ Serotonin increase and be able to transmit impulses at the synapses as levels are high enough for Serotonin to be used as a neurotransmitter as low levels inhibit the transmission of impulses. ^{This allows} ~~the transmission of~~ ^{to be passed} an impulse ~~to the~~ to the post synaptic axon, meaning that the level of depression will be reduced due ~~to~~ the increase of Serotonin. Other medication may inhibit the absorption of Serotonin. Serotonin is used as a neurotransmitter when an action potential passes through the synapses. Therefore when it is ^{passed} ~~fast~~ through the receptors of a post synaptic neurone it isn't ~~reabsorbed~~ but recycled back to the presynaptic axon to be used again. This stops levels of ^{Serotonin} ~~Serotonin~~ from decreasing, ~~and~~ therefore preventing depression as Serotonin levels are maintained.

(Total for Question 5 = 6 marks)

The next response was awarded 2 marks. The learner discussed their idea of mimicking the usual transmission at a faster rate, but there are no details

about being an agonist or filling the receptors and so this was a level 1 response.

Discuss different ways prescribed drugs could act at a synapse to treat depression.

Use the information in Figure 6 to support your answer.

(6)

- The prescribed drugs could trigger the increase of serotonin levels as the synapse have vesicles ~~store~~ storing serotonin and they are released through electrical impulses and attached to receptors. These receptors then transmission signals to your brain, making you feel happier and relaxed.
- the drugs in return can mimic this process at a faster rate and possibly in higher ratios as you can be prescribed stronger doses which will increase serotonin levels and then maintain them.

Chemistry

Question 1

The first question on the paper focused on water. **1(a)** was a multiple-choice question, learners were asked to determine the relative molecular mass of water. Around 76% were able to select 18 as the correct answer.

1(b) asked learners to state the type of bond between the hydrogens atoms and the oxygen atom in Figure 1. Over half of learners gained the mark for understanding that it is a covalent bond as in this example.

(b) State the type of bond between the hydrogen atoms and the oxygen atom in Figure 1.

(1)

Covalent bond

Of those that did not score, a common error was where learners did not read the question carefully, which asked for the type of bond between the atoms

and stated that it was a hydrogen bond as in this example that did not gain credit. A smaller proportion thought that it was an ionic bond.

(b) State the type of bond between the hydrogen atoms and the oxygen atom in Figure 1.

(1)

Hydrogen bond

1(c) – Learners found this second multiple-choice question harder with around 14% understanding that water would contains all three of the types of intermolecular force.

1(d) gave learners a table showing the boiling points of hydrogen and of oxygen. They were asked to explain why the boiling point of hydrogen is lower than the boiling point of oxygen. Learners found this quite challenging, with many learners referring to bonds and what affects the strength of bonds, rather than intermolecular forces. This confusion led some learners to talk about the impact of the number of electrons on the outer shell, rather than the number of electrons in the entire molecule.

This example scored 1 mark for stating that hydrogen has fewer electrons. Unfortunately, they did not score the second mark as they refer to the energy to break the bond between the electrons and the nucleus rather than the intermolecular force.

Table 1 shows the boiling point of hydrogen, H_2 and oxygen O_2 .

	boiling point ($^{\circ}C$)
hydrogen	-253
oxygen	-183

Table 1

(d) Explain why the boiling point of hydrogen is lower than the boiling point of oxygen.

(2)

hydrogen has fewer electrons and only one shell so the amount of energy needed would be lower due to a weaker bond between electrons and nucleus.

(Total for Question 1 = 5 marks)

Exam technique was an issue here and with some learners losing marks because they did not compare hydrogen to oxygen sufficiently for example stating that hydrogen had weak intermolecular forces rather than saying weaker intermolecular forces than oxygen. Others lost the second mark as they were not careful and specific with their use of key terms, for example stating that less energy was needed to break bonds rather than break intermolecular forces or intermolecular bonds.

Question 2

2(a)(i) asked learners about the reaction of magnesium with hydrochloric acid. They were given the products of the reaction and asked what would be observed in the boiling tube. In general, the question was answered well with a good proportion knowing that when magnesium is added to hydrochloric acid, fizzing would be observed due to the release of hydrogen, few referred to the test tube getting warmer. Of those that did not score, it was often as they did not read the question carefully to note that it was about observations, and so repeated the stem, stating that hydrogen or a gas was formed, these ideas were ignored. Others gave incorrect observations such as a colour change or sparks so did not score a mark. In some cases, learners were not careful with their key terms and stated that the magnesium would dissolve this was ignored.

This example gained the mark.

2 A learner places a small piece of magnesium into a boiling tube containing 25 cm³ of dilute hydrochloric acid, HCl.

Magnesium chloride and hydrogen are produced.

(a) (i) State what would be observed in the boiling tube when the magnesium is added to the dilute hydrochloric acid.

(1)

Fizzing

This next example gained no marks, although the learner had stated that bubbles would be formed, this was then contradicted by stating that a flame would be seen. Learners should be taught not to give lists of answers.

- 2 A learner places a small piece of magnesium into a boiling tube containing 25 cm³ of dilute hydrochloric acid, HCl.

Magnesium chloride and hydrogen are produced.

- (a) (i) State what would be observed in the boiling tube when the magnesium is added to the dilute hydrochloric acid.

~~Shiny substance~~ (1) Bubbling and a light flame

A number of responses stated that hydrogen would be formed, which suggests an error in interpreting the question as only observations could be credited. The most common incorrect response involved the formation of smoke, the production of a flame or a popping sound being produced.

2(a)(ii) asked learners for the balanced equation for the reaction, Learners found writing the balanced equation difficult, with many being unable to give the correct formula for magnesium chloride or knowing that hydrogen is diatomic. In some cases, learners did not give the correct formula for hydrochloric acid, even though this was given in the stem of the question. Learners should be taught to use the stem of the question to help inform their answers. Of those that were able to correctly write the molecular notation for both MgCl₂ and H₂, the majority were then able to go on to balance the equation correctly to gain both marks. In a smaller number of cases, learners did not read the question carefully and tried to give the word equation rather than the balanced symbol equation, this gained no credit. Learners should be taught the difference between a word and a balanced symbol equation and when each should be used.

It was very common to see learners giving the formula of magnesium chloride as MgCl, and that formula for hydrogen as H, as in this example which scored 0 marks. As the formula was not correct, the balancing mark could not be awarded.

- (ii) Write the balanced symbol equation for this reaction.

(2)



This example shows a good answer that scored both marks.

(ii) Write the balanced symbol equation for this reaction.

(2)



2(b)(i) asked learners to calculate the number of moles of hydrochloric acid in a sample of hydrochloric acid, they then used this to calculate the molar concentration of the sample in part (ii).

It was pleasing to see that the majority of learners were able to divide the mass given in the question by the relative formula mass to gain the first mark. In some cases, the learners inverted the division and so lost the mark. Error carried forward was however allowed so if they used this value correctly in part (ii) they were able to achieve full marks for part (ii).

2(b)(ii) – Some learners struggled with converting cm³ to dm³, often incorrectly multiplying instead of dividing by 100 instead of dividing by 1000. A large proportion of learners didn't attempt any conversion and so lost a mark.

In this example the learner has correctly calculated the number of moles in 2(b)(i). In 2(b)(ii) they have divided by 100 rather than 1000 so lost the conversion mark. Error carried forward was allowed and the learner gained 2 marks for the correct calculation of molar concentration without the conversion and evaluation.

(b) A sample of dilute hydrochloric acid contains 1.825 g of hydrochloric acid, HCl.

(i) Calculate the number of moles of hydrochloric acid in the sample.

(relative formula mass: HCl = 36.5)

$$\frac{1.825}{36.5} = 0.05 \quad \text{Mr} \quad m \quad n \quad (1)$$

number of moles of HCl = 0.05

(ii) Calculate the molar concentration of 25 cm³ of the dilute hydrochloric acid sample.

Use your answer from part (b)(i) in your calculation.

If you did not get an answer for part (b)(i) use 0.07 for the number of moles.

(3)

Show your working.

$$\begin{array}{r} 25 \text{ cm}^3 \\ \div 100 \\ \hline 0.25 \end{array} \quad \text{c} \quad \text{v} \quad n$$

$$\frac{0.05}{0.25} = 0.2$$

molar concentration = 0.2 mol dm⁻³

In the next example, the learner has incorrectly calculated the number of moles in 2(b)(i) as 0.025 and so lost the mark. Error carried forward was allowed and the learner gained all three marks for correctly converting the volume, calculating the volume and evaluating their answer correctly using this value.

(b) A sample of dilute hydrochloric acid contains 1.825 g of hydrochloric acid, HCl.

(i) Calculate the number of moles of hydrochloric acid in the sample.

(Mr) (relative formula mass: HCl = 36.5)

$$\frac{1 \times 36.5}{1 \times 36.5} = 36.5$$


$$\frac{1.825}{36.5} = 0.025 \quad (1)$$

number of moles of HCl = 0.025

(ii) Calculate the molar concentration of 25 cm³ of the dilute hydrochloric acid sample.

Use your answer from part (b)(i) in your calculation.

If you did not get an answer for part (b)(i) use 0.07 for the number of moles.

(3)

Show your working.



$$\frac{0.025}{25 \text{ cm}^3} = \frac{0.025}{0.025 \text{ dm}^3}$$

÷ 1000

0.025,000

molar concentration = 1 mol dm⁻³

The next response shows a good answer – showing full, logical working to gain full marks on part (i) and part (ii). Whilst working is not compulsory to gain the full marks if the answer is correct, is important that learners do show their workings in a systematic way to help them form their answer and so that part marks and error carried forward can be awarded.

(b) A sample of dilute hydrochloric acid contains 1.825 g of hydrochloric acid, HCl.

(i) Calculate the number of moles of hydrochloric acid in the sample.

(relative formula mass: HCl = 36.5)

$$\text{Moles} = \frac{\text{Mass}}{\text{molar mass}}$$

$$\frac{1.825}{36.5} \quad (1)$$

$$\text{number of moles of HCl} = \frac{0.05}{0.05} \quad 0.05$$

(ii) Calculate the molar concentration of 25 cm³ of the dilute hydrochloric acid sample.

Use your answer from part (b)(i) in your calculation.

If you did not get an answer for part (b)(i) use 0.07 for the number of moles.

(3)

Show your working.

$$\text{Concentration} = \frac{\text{Moles}}{\text{Volume}}$$

$$\frac{0.05}{0.025}$$

$$\text{Moles} = 0.05$$

$$\text{Volume} = 25 \text{ cm}^3$$

$$\frac{25 \text{ cm}^3}{1000} = 0.025 \text{ dm}^3$$

$$1000$$

$$\text{molar concentration} = 2 \text{ mol dm}^{-3}$$

In some cases, where learners were unsure of their answer for the number of moles to 2(b)(i) they used the given value in part (b)(ii) which was acceptable. In the next example, the learner has multiplied the mass given by the relative formula mass so lost the mark in 2(b)(i). In 2(b)(ii) they have used the given value to correctly calculate the molar concentration and was awarded the full 3 marks.

(b) A sample of dilute hydrochloric acid contains 1.825 g of hydrochloric acid, HCl.

(i) Calculate the number of moles of hydrochloric acid in the sample.

(relative formula mass: HCl = 36.5)

$$\frac{\text{moles}}{\text{nr T mass}} \quad 36.5 \times 1.825 = 66.6125 \quad (1)$$

$$\text{number of moles of HCl} = 66.6125$$

(ii) Calculate the molar concentration of 25 cm³ of the dilute hydrochloric acid sample.

Use your answer from part (b)(i) in your calculation.

If you did not get an answer for part (b)(i) use 0.07 for the number of moles.

Show your working.

$$\frac{25}{1000} = 0.025 \quad (3)$$

$$\frac{0.07}{0.025} = 2.8$$

$$\text{molar concentration} = 2.8 \text{ mol dm}^{-3}$$

2(c) asked learners to explain the difference in the reaction when sodium, rather than magnesium, is reacted with hydrochloric acid. A large proportion of learners thought that magnesium was more reactive than sodium, as in this example that scored no marks.

The learner watched a video of the reaction of sodium metal with dilute hydrochloric acid.

- (c) Explain the difference in the reaction when sodium, rather than magnesium, is reacted with dilute hydrochloric acid.

(2)

Magnesium will show more of a reaction than sodium because Magnesium is a more reactive metal

Of those that did understand that sodium is more reactive than magnesium, few were able to explain why sodium is more reactive to gain the second mark, whilst some learners correctly stated the difference in the number of electrons (on the outer shell) they failed to recognise that the electron was needed to be lost.

This example gained 1 mark for describing a more vigorous reaction with sodium, there was no reason for why there is more fizzing. Learners should be taught that when a question asks them to explain they need to make their point and then given a justification of exemplification of the point, the answer must contain some reasoning.

The learner watched a video of the reaction of sodium metal with dilute hydrochloric acid.

- (c) Explain the difference in the reaction when sodium, rather than magnesium, is reacted with dilute hydrochloric acid.

(2)

When sodium is used instead of magnesium then there is more fizzing than with magnesium as sodium is a group 1 element which is very reactive.

(Total for Question 2 = 9 marks)

This example shows a good answer which gained both marks.

The learner watched a video of the reaction of sodium metal with dilute hydrochloric acid.

- (c) Explain the difference in the reaction when sodium, rather than magnesium, is reacted with dilute hydrochloric acid.

(2)

Sodium is more reactive than magnesium because it requires less energy due to only losing one electron to become an ~~ion~~ ion where as magnesium needs to lose two. This means ~~the~~ reaction with sodium will cause more fizzing than the one with magnesium

(Total for Question 2 = 9 marks)

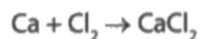
Question 3

This question focused on calcium and its compounds.

3(a) was a multiple choice question which asked learners to identify which orbital contains calcium's outermost electrons. 54% of learners were able to correctly select 4s as the correct orbital.

3(b)(i) – Learners were given the balanced equation of calcium with chlorine to form calcium chloride. They were told that calcium chloride is a solid ionic compound and asked to give two physical properties of calcium chloride. A large proportion of learners did not read the question carefully and thought that calcium chloride was a metal rather than an ionic compound and so gave properties of a metal such as malleable, ductile, good conductor of heat or can conduct electricity, without referring to being in aqueous, molten or as a liquid as in this example which scored no marks. If they had stated that it could conduct when molten or as a solution, the mark could have been awarded. Learners should be taught to be precise with their answers so that there is no ambiguity and marks can be awarded.

(b) Calcium reacts with chlorine to form the solid ionic compound calcium chloride.



(i) Give **two physical** properties of calcium chloride.

(2)

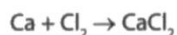
it ~~has~~ is malleable

it can conduct electricity

Another common error seen was to describe the physical arrangement (lattice)/state of matter (solid), rather than physical properties. Some learners did not read the question properly or did not understand what is meant by the term property as they proposed uses of calcium chloride rather than properties. Of those that did score, it was often as they stated that calcium chloride had either a high boiling point or high melting point.

This example scored 1 mark, although the learner has stated high melting point and high boiling point, these were the same marking point.

(b) Calcium reacts with chlorine to form the solid ionic compound calcium chloride.



(i) Give **two physical** properties of calcium chloride.

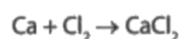
(2)

1 ~~Soft~~ high melting point

2 ~~Ductile~~ ~~Strong~~ high boiling point

This example shows a good answer that scored both marks.

(b) Calcium reacts with chlorine to form the solid ionic compound calcium chloride.



(i) Give **two physical** properties of calcium chloride.

(2)

- 1 soluble in water
- 2 with ionic bonds high melting point

3(b)(ii) – Learners were reminded that calcium and chlorine bond ionically to form calcium chloride and gave learners a diagram showing the outer electrons in an atom of calcium and an atom of chlorine, they were then asked to draw a dot and cross diagram for calcium chloride. They were told that they must show the charges on the ions. Learners found this quite challenging with over half scoring no marks. Even though mentioned the stem that the bond was ionic and to draw charges on the ions, many learners still attempted to share electrons and covalent bonds which scored no marks as in this example:

Calcium and chlorine bond ionically to form calcium chloride.

Figure 2 shows the outer electrons in an atom of calcium and an atom of chlorine.



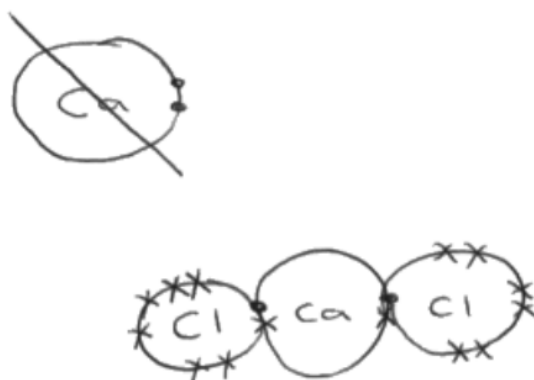
Figure 2

(ii) Draw a dot and cross diagram for calcium chloride.

Use the information in Figure 2 to help you.

You must show the charges on the ions.

(3)



In some cases, whilst the chloride ions were drawn correctly. The mark couldn't then be awarded for the calcium ion as the electron configuration was not in its final form. Where learners understood how to draw the dot and cross diagrams correctly, they often failed to show both outer electrons being transferred from calcium and drew just one chloride ion so only gained one mark for one correctly drawn ion and the calcium shown with one outer electron and a 1+ charge as in this example that scored 1 mark for the one chloride ion only.

Calcium and chlorine bond ionically to form calcium chloride.

Figure 2 shows the outer electrons in an atom of calcium and an atom of chlorine.

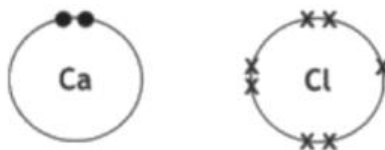


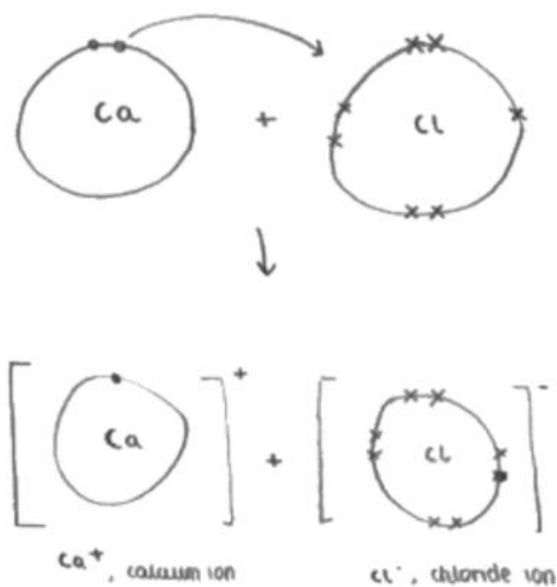
Figure 2

(ii) Draw a dot and cross diagram for calcium chloride.

Use the information in Figure 2 to help you.

You must show the charges on the ions.

(3)



This example shows a good answer that scored both marks.

Calcium and chlorine bond ionically to form calcium chloride.

Figure 2 shows the outer electrons in an atom of calcium and an atom of chlorine.

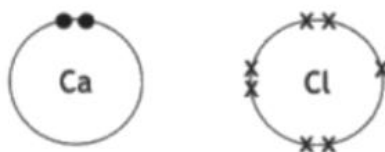


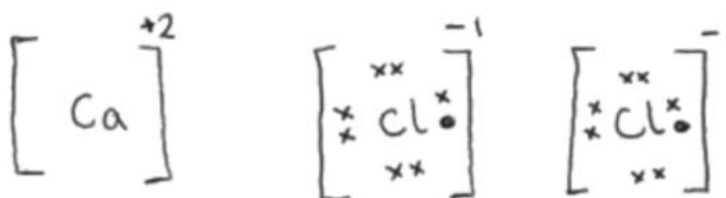
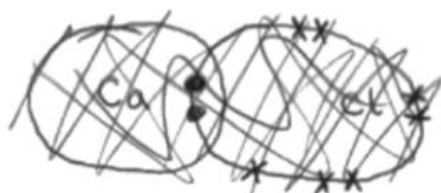
Figure 2

(ii) Draw a dot and cross diagram for calcium chloride.

Use the information in Figure 2 to help you.

You must show the charges on the ions.

(3)



Q3(b)(iii) – Learners were given the equation for the reaction again and asked to explain why the reaction between calcium and chlorine is a redox reaction. They were told that they must refer to changes in oxidation number in their answer. Learners found the question challenging with a very small proportion scoring the 4 marks available.

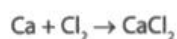
Where learners did score it was largely as they referred to the gain and loss of electrons on chlorine and calcium respectively, or for stating that calcium is oxidised, and chlorine is reduced. Where learners correctly identified the new oxidation number, they often failed to state the original oxidation number so didn't get any credit.

Learners that did not score often did so as they gave generic answers rather than explaining why this reaction specifically is a redox reaction, regularly

stating that reduction and oxidation happened but not specifying which one was which.

Some learners were confused in their answers and mixed their definitions as in this example, where the learner states that calcium is oxidised which would have gained the mark, but unfortunately, they state that the calcium gains an electron which is wrong and so negated the mark.

The reaction of calcium with chlorine is a redox reaction.



(iii) Explain why the reaction of calcium with chlorine is a redox reaction.

You must refer to changes in oxidation number in your answer.

(4)

Handwritten student answer on lined paper:

Ca —————> Ca
0 —————> +1

~~Cl —————> Cl~~
~~0 —————> +2~~

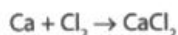
Calcium gains an electron so it's oxidised

Chlorine gains 2 electrons so it's reduced

(Total for Question 3 = 10 marks)

A common error seen of those that did answer well is exemplified in this answer. The learner explains that calcium has been oxidised as it has lost electrons and give the correct change in oxidation number. They state that chlorine has been reduced as it gains electrons, however they give the oxidation number as -2 rather than -1 and so loses the final mark.

The reaction of calcium with chlorine is a redox reaction.



(iii) Explain why the reaction of calcium with chlorine is a redox reaction.

You must refer to changes in oxidation number in your answer.

(4)

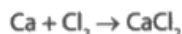
Calcium ~~has~~ begins with an oxidation number of 0. It then has an oxidation number of +2 so it has lost ~~the~~ 2 electrons so it has been oxidised.

Chlorine begins with an oxidation number of 0. It then has an oxidation number of -2 therefore it has gained two electrons so it has ~~been~~ been reduced.

(Total for Question 3 = 10 marks)

This example shows a good answer that scored the 4 marks available.

The reaction of calcium with chlorine is a redox reaction.



(iii) Explain why the reaction of calcium with chlorine is a redox reaction.

You must refer to changes in oxidation number in your answer.

(4)

Calcium has been oxidised as it has given up 2 electrons to chlorine. Calcium has gone from 0 to 2^+ . This makes it a REDOX reaction as chlorine is also then reduced, it ~~takes~~ gains one electron per chlorine, taking it from 0 to 1^- (-) per chlorine.

(Total for Question 3 = 10 marks)

Question 4

The last question on the paper gave learners the electronegativity values of some elements in the periodic table. They were asked to discuss the trends in electronegativity values shown in the figure and to include why there are no values for the noble gases.

Many learners were able to give the trends in electronegativity across periods and down groups to gain some marks, however many were confused about the definition and often discussed the ability to lose and electrons and ionization energy rather than electronegativity.

A good proportion knew that noble gases had a full outer shell of electrons however were not able to explain in any more detail why this meant that they had no electronegativity values. A noticeable proportion thought that they had no electrons and so therefore no electronegativity values.

Few learners were able to give comprehensive analysis, linking more than one factor to explain a trend. Of those that did, most were able to link the number of shells, shielding and to nuclear attraction to the incoming pair. Trends across periods were less well known. Many learners confused electronegativity with factors effecting ionisation, or explained the absolute electronegativity of one element, usually F, rather than a trend. Some learners were able to link the increased reactivity across the period to the increased nuclear charge and decreased atomic radius. However, some learners linked it to the increase in electrons in the outer shell.

The decreased electronegativity was often linked to the increased atomic radius. However, learners did not always include the increased nuclear charge and shielding in their answers.

This example gained 2 marks in level 1. The learner has given a basic definition of electronegativity and some factors that affect it. They have stated the trends across the periods and down the groups and basic explanation of why the noble gases have no values.

4. Figure 3 shows the elements in the first three periods of the periodic table.

The electronegativity values of the elements are shown in Figure 3.

There are no values for the noble gases.

H							He
2.1							—
Li	Be	B	C	N	O	F	Ne
1.0	1.5	2.0	2.5	3.0	3.5	4.0	—
Na	Mg	Al	Si	P	S	Cl	Ar
0.9	1.2	1.5	1.8	2.2	2.5	3.0	—

Figure 3

Discuss the trends in electronegativity values shown in Figure 3.

Your answer should include why there are no values for the noble gases.

(6)

Electronegativity is the attraction that the electrons have to their nucleus.

It is affected by the number of protons, radius of the element/molecule and no. of electron & no. of electrons on the outer shell.

Noble Gases have full and complete outer shells therefore, would never lose or gain an electron.

Electronegativity increases left to right of the periods and Bottom to top in the groups.

In the next example, the learner states that the electronegativity increases across a period due to an increase in protons, that the electronegativity decreases down a group due to more shielding and that the noble gases have a full outer shell, unfortunately they link this to attracting an electron from another atom - implying ionic bonding rather than electronegativity. As the learner has given basic explanations only, but of the three trends the answer gained a mark of 4 in level 2.

4 Figure 3 shows the elements in the first three periods of the periodic table.

The electronegativity values of the elements are shown in Figure 3.

There are no values for the noble gases.

H 2.1							He —
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne —
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.2	S 2.5	Cl 3.0	Ar —

Figure 3

Discuss the trends in electronegativity values shown in Figure 3.

Your answer should include why there are no values for the noble gases.

(6)

Electronegativity increases across a period due to an increase in protons and thus charge. electronegativity decreases down a group due to the atoms getting larger and having more shielding. There are no units for the noble gases because they have a full outer shell and do not need to attract an outer shell electron from another atom.

Electronegativity is the atoms ability to attract an outer shell electron from another molecule, this explains why the noble gases have no unit.

In the next example, the learner has explained the increasing electronegativity across the period and the decreasing electronegativity down the groups. They explain why the noble gases do not have values so and so therefore gained full marks in level 3.

4 Figure 3 shows the elements in the first three periods of the periodic table.

The electronegativity values of the elements are shown in Figure 3.

There are no values for the noble gases.

H 2.1							He ---
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne ---
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.2	S 2.5	Cl 3.0	Ar ---

Figure 3

Discuss the trends in electronegativity values shown in Figure 3.

Your answer should include why there are no values for the noble gases.

(6)

Electronegativity is the relative power of an atom to attract the shared pair of electrons towards itself within a covalent bond

The most electronegative elements are fluorine, oxygen & chlorine

Electronegativity depends on: nuclear charge

shielding

atomic radius

As you go across the period from lithium to fluorine electronegativity increases, this is because nuclear charge increases, whilst shielding stays the same, this means atomic radius decreases as there is a stronger electrostatic attraction between the nucleus & outer electron, increasing electronegativity

As you go down a group for example from hydrogen to sodium (group 1) electronegativity decreases, this is because as you go down a

group number of shells increases, this means the effect of shielding is greater whilst also increasing the atomic radius. As a result the attraction between the nucleus & outer electrons is weaker.

Noble gases do not have values because they have a full outer shell & are completely unreactive. This means they can not ^{as they don't need to share electrons} form a covalent bond and ~~therefore cannot~~ ~~have electronegativity values~~ therefore cannot have electronegativity values.

* shown as lithium's electronegativity is 1.0 while fluorine is 4.0

* shown as hydrogen's electronegativity is 2.1 while sodium is 0.9

Physics

Question 1

1(a) – Most learners were able to recall that all electromagnetic waves have the same speed in a vacuum.

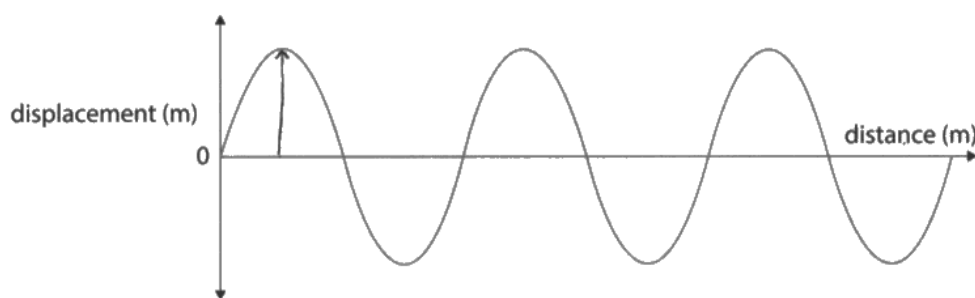
(a) Identify which feature is the same for all electromagnetic waves in a vacuum.

(1)

- ☐ A amplitude
- ☐ B frequency
- ☐ C speed
- ☐ D wavelength

1(b) – Learners were provided with the diagram shown below and asked to draw an arrow to show the amplitude of the wave.

The majority of learners were able to do so correctly.



(Source: © PAL)

1(c) – Many learners struggled to correctly identify option A as being the correct response to the question below showing a ray of light travelling from glass into air. A common mistake was to select option C, suggesting difficulty in interpreting angles of incidence and refraction from a ray diagram.

Figure 2 shows a ray of light refracting as it travels from glass into air.

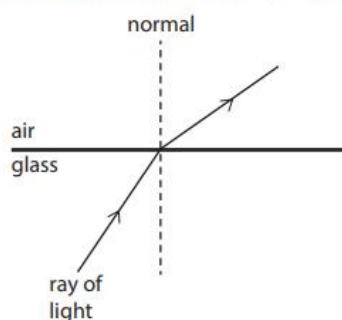


Figure 2

(c) Which row of the table shows the angles of incidence and refraction in Figure 2?

(1)

	angle of incidence (degrees)	angle of refraction (degrees)
☐ A	33	55
☐ B	20	13
☐ C	55	33
☐ D	13	20

1(d) – The majority of learners were able to complete most of the calculation using the equation provided. The most common mistake was to forget to perform the step to derive the square root. The following response gained full marks.

A transmitter sends a microwave signal to a mobile phone.

The intensity (I) of the signal received by the mobile phone is $2.2 \times 10^{-5} \text{ W m}^{-2}$.

(d) Calculate the distance (r) from the transmitter to the mobile phone.

(4)

Use the equation

$$I = \frac{k}{r^2}$$

The constant k is 1.5 W .

Show your working.

$$2.2 \times 10^{-5} = \frac{1.5}{r^2}$$

$$\sqrt{68181.81} \\ = 261.1$$

$$2.2 \times 10^{-5} \times r^2 = 1.5$$

$$\frac{1.5}{2.2 \times 10^{-5}} = 68181.81$$

$$\text{distance} = 261.1 \text{ m}$$

Centres should encourage learners to show their workings when completing calculations, as many can gain marks for the process used, even if their final answer is incorrect.

Question 2

2(a) – Some learners were able to suggest a reason why infrared is suitable for TV remote controls, with the idea of it able to act over the short distance between a TV and the control being the most popular response seen.

2 Electronic devices can use infrared or Bluetooth® technology.

Remote controls communicate with televisions using infrared technology.

(a) State **one** reason why infrared technology is suitable for remote controls.

(1)

It has a short wavelength so can be used over short distances needed for a remote control

2(b) – Whilst the majority of learners could suggest a disadvantage of using Bluetooth® technology, only a minority were able to explain the reason for this type of technology having the disadvantage identified.

Some learners suggested a disadvantage of the technology but then gave an explanation of a different disadvantage- centres should remind learners that for these types of questions, the explanation has to be linked to the identification for both marks to be awarded. The following response only gained 1 mark as the identification and the explanation were not a linked pair.

(b) Explain **one** disadvantage of using Bluetooth® technology to communicate between electronic devices.

(2)

Bluetooth has a short band width, so less data is transferred. Other bluetooth devices may interfere if close by and on similar frequencies.

2(c) Many learners were able to use the equation $v = f \times \lambda$ to perform most steps of the calculation correctly, gaining full at least 3 marks. An example gaining full marks is shown below:

The wavelength (λ) of an infrared wave is 800 nm.

The wave has a velocity (v) of $3.0 \times 10^8 \text{ m s}^{-1}$.

(c) Calculate the frequency (f) of the wave.

(4)

Use the equation

$$v = f \times \lambda$$

Show your working.



$$800 \text{ nm} \rightarrow \times 10^{-9}$$

$$f = \frac{3.0 \times 10^8}{800 \times 10^{-9}} = 3.75 \times 10^{14}$$

frequency = 3.75×10^{14}

frequency = ~~3.75~~ 3.75×10^{14} Hz

Whilst the correct answer would be awarded full marks even if workings are not shown, centres are recommended to encourage learners to include them, as marks can be awarded for the correct process steps, even if the final answer is incorrect. An example of this is shown in the following response.

The wavelength (λ) of an infrared wave is 800 nm.

The wave has a velocity (v) of $3.0 \times 10^8 \text{ m s}^{-1}$.

$$800 \text{ nm} \rightarrow \text{m} \\ = 0.80$$

(c) Calculate the frequency (f) of the wave.

(4)

Use the equation

Show your working.

$$v = f \times \lambda \\ 3.0 \times 10^8 = f \times 0.80 \\ \frac{3.0 \times 10^8}{0.80} = f$$

$$\text{frequency} = 375000000 \text{ Hz}$$

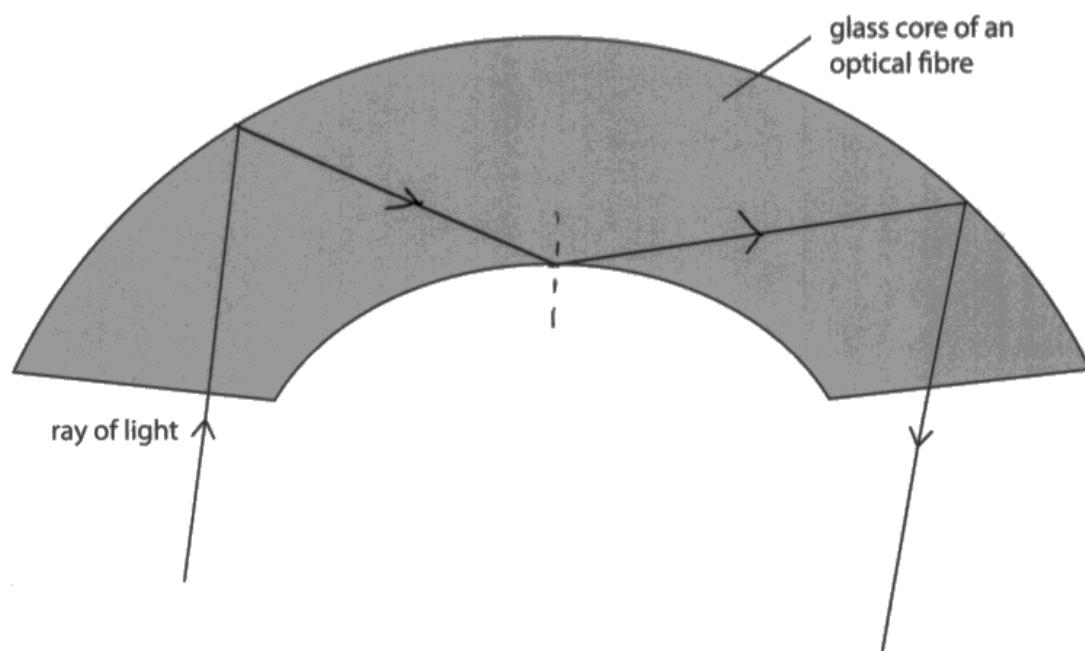
The workings clearly show the learner's thought process in performing this calculation. It can be seen that there is an error in the conversion step (the most common error seen) but MP2 and MP3 can still be awarded as substitution and rearrangement are correct (error carried forward from incorrect conversion). Additionally, MP4 can also be awarded as the evaluation is correct for values used (error carried forward).

Question 3

3(a) – Many learners were able to correctly complete the diagram to show the path a light ray would take through an optical fibre. The following response gained 1 mark.

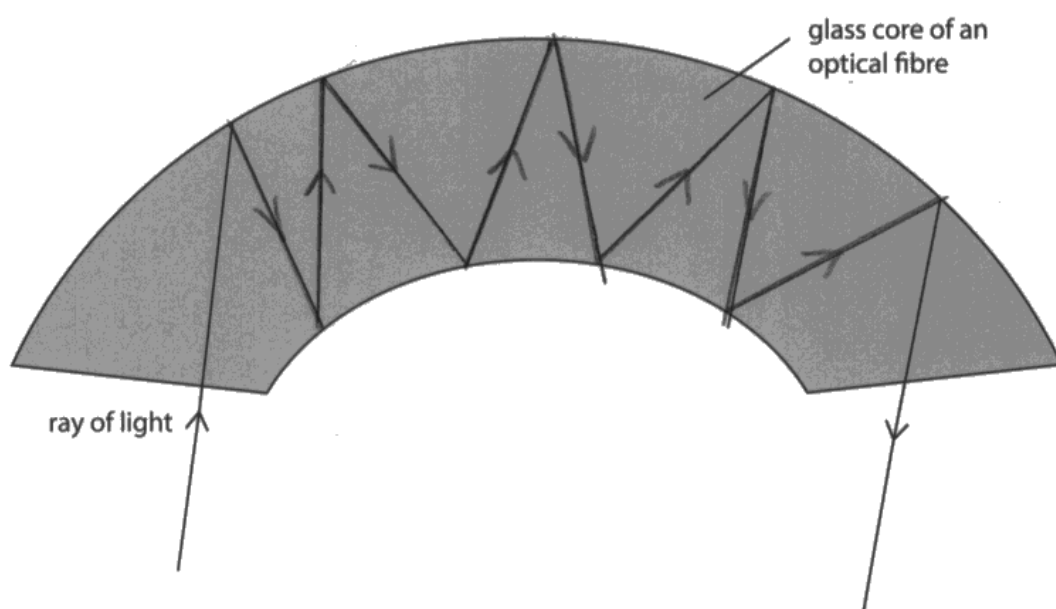
Endoscopes use optical fibres.

Figure 4 shows a ray of light entering and leaving the glass core of an optical fibre.



Centres should remind learners that whilst the use of a ruler and protractor are not required, care should be taken in drawing ray diagrams, with consideration given to show the path taken. The following response shows the ray escaping from the glass core, and the reflections drawn are not accurate (even by eye) in terms of the angles of incidence and reflection. As a result, this response did not gain the mark available.

Figure 4 shows a ray of light entering and leaving the glass core of an optical fibre.



3(b) – This question required learners to correctly recall the term used to describe the process by which a ray of light travels through an optical fibre.

Many learners struggled to complete this correctly, and either left it blank or used words related to the behaviour of light, for example, 'refracted diffracted'. Learners should be encouraged to read over their answers and check for sense.

(b) Complete the sentence by adding **two** words.

(1)

The ray of light travels through the optical fibre by a process called

Total Internal reflection.

3(c) – Many learners were unable to suggest a benefit of using optical fibres to see inside a patient, with the most common correct response being that it provided an alternative to surgery.

A number of learners described the process of endoscopy as being 'harmless' or that it provided a clear image without further qualification. An example of this is shown in the response below.

An endoscope can be used to see inside a patient's stomach.

(c) State **one** benefit of using optical fibres to see inside the patient's stomach.

(1)

gives a clear image

This was not sufficient to gain the mark, as it could be argued that other techniques, including surgery, would also allow clarity of visualising what was happening within the patient.

3(d) – This question required learners to demonstrate understanding of how cladding in an optical fibre minimises light loss. The following response gained full marks and demonstrates good knowledge of the topic.

The structure of an optical fibre is a glass core surrounded by outer cladding.

(d) Explain how the structure of an optical fibre minimises the amount of light lost from the glass core.

(3)

- The cladding ~~increases~~ has a lower refractive index
- This increases the critical angle and means fewer reflections occur
- ~~Therefore~~ ^{less} light energy is lost as ~~the~~ it doesn't have to travel as much as in a distance

Many learners struggled to explain how this happens, and a significant number of learners completely misunderstood the role of cladding in an optical fibre, suggesting it gave some protection to stop the glass core breaking or was dark in colour to stop light escaping. Some learners simply repeated the information given in the question stem as the following response shows.

The structure of an optical fibre is a glass core surrounded by outer cladding.

(d) Explain how the structure of an optical fibre minimises the amount of light lost from the glass core.

(3)

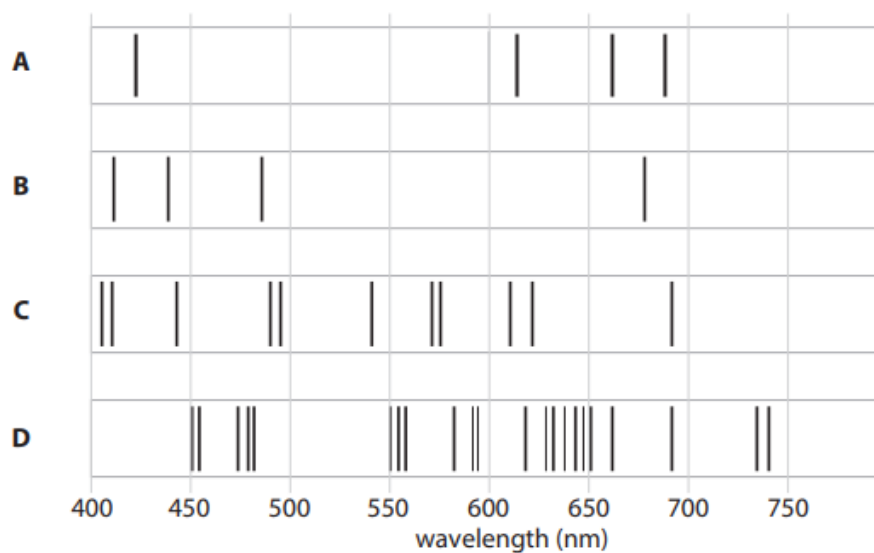
Surrounding the glass core there is cladding - Cladding surrounds the glass core and keeps the light in.

This response gained no marks.

Question 4

4(a) – The majority of learners were able to correctly identify the emission spectrum for mercury, choosing option C as their response.

4 Figure 5 shows four emission spectra.



(Source: © PAL)

Figure 5

The emission spectrum for the element mercury has lines at 405 nm and 546 nm.

(a) Identify the emission spectrum for mercury.

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D

4(b) – Many learners found this question very difficult, indicating limited understanding of the significance of the lines found in emission spectra. The following response gained full marks.

The emission spectrum is produced after atoms of mercury have absorbed energy.

(b) Explain why an emission spectrum consists of a series of lines.

(3)

Energy is absorbed by electrons in a sample. These electrons excite and move to a higher energy level. In this state, they are unstable, so eventually fall back (de-excite) to their original energy level, and in doing so emit photons of light of a specific wavelength, which is recorded on an emission spectra, as specific lines as seen above.

(Total for Question 4 = 4 marks)

A common error seen was in describing atoms being excited, rather than electrons. There were also attempts made to describe applications of emission spectra such as the analysis of stars. Centres should remind students to check that their response answers the question asked.

An example of this is shown below.

The emission spectrum is produced after atoms of mercury have absorbed energy.

(b) Explain why an emission spectrum consists of a series of lines.

(3)

When atoms get "excited" they they emit a multitude of colours each colour has its own wavelength which can be determined with colours such as red being smaller than indigo/violet and these colours are used to identify elements in things such as stars

Question 5

This question required learners to interpret trends on a graph and then relate them to their knowledge of analogue and digital signals. Many learners struggled to do both, with some describing the graph in great detail, but not using their knowledge to explain the trends shown. Other learners discussed analogue and digital signals, but made no or very limited reference to the graph.

The following response met the level 1 descriptors as it consists of a series of statements about analogue and digital signals, with no linkage. It was awarded 2 marks.

Discuss the quality of analogue signals and digital signals.

Include in your answer how quality varies with strength.

Use the information in Figure 6 to support your answer.

(6)

- Digital can be regenerated
- travels further without degradation
- Analogue Picks up unwanted noise, more interference, worse quality of signal
- Digital can get perfect signal.
- Digital has no loss of info.
- ~~Digital~~ is stronger signal
analogue
- Digital is a discontinuous system of only two values, 0 and 1.
- Analogue is a continuous signal with more than two values.
- ~~analogue takes longer to pick up~~
- analogue takes gradual time to pick up a strong signal, digital is fast and instant.
- unwanted sound can be removed from digital signal.
- analogue doesn't know difference between signal and sound.

The next response met the level 2 descriptors and was awarded 4 marks. Both types of signal are discussed, with some linkage of ideas demonstrated. There is an attempt to relate aspects of the graph to knowledge of analogue and digital signals.

The quality of analog and digital signals are mainly varied, with factors like the what method is being used to transfer the signals. Analog Digital signals are known to have more of a better lossless quality than analog signals, most likely due to the fact that it is usually not prone to noise, unlike analog signal. When noise affects a signal, the quality heavily decreases, but and that might not let the right information be processed. Also, if a signal is filled with noise or affected by it, it may not be a strong signal anymore. This can mainly affect analogue signals as they are not stored in 1s and 0s but have ~~analog~~ digital signals, but rather in variables. This means ~~that~~ not that the quality also depends on the variables sent in the analog. The graph shows digital signals having an immediate change in both signal strength and signal quality while analogue signals ~~are~~ have a gradient even though it also reaches 100% signal quality. The gradient shows its ~~quicker~~ slow climb between 0 to 40% signal strength then a rise to which then shows down at around 70% but still reaches 100%. ~~This is rather slow than some~~ This therefore means that the signal ^{quality} ~~strength~~ is dependent on signal strength. This, compared to ~~analog~~ digital ^{signals} ~~quality~~ on which has no response within a low signal strength but has high immediately reaches 100% at a quicker pace may outline that digital signals with output better quality faster and better than signal strength.

The next response met the level 3 descriptors and was awarded six marks. This response makes a number of correct comments about both types of signal, demonstrating a good understanding of this topic, and linking ideas together e.g. 'over a greater distance they can be regenerated so no signal is

lost'. The response describes the trends shown in the graph (there is an error in the comment about analogue reaching 100% quality at 60% strength, but otherwise the rest are correct). The response also links ideas to explain one of the trends in the graph, for example, 'signal quality is 100% as this is when it is on'. The answer shows a well-developed structure which is clear, logical, and coherent.

(6)
- Analogue signals vary with amplitude and frequency. They can also easily be interfered with, they require more bandwidth and over a greater distance the signal can be lost. As signal quality increases so does the signal strength until 60% signal strength where it then becomes 100% signal quality as the signal strength increases.

- Digital signals is binary values. They won't be interfered by noise and require less bandwidth. And over a greater distance they can be regenerated so no signal is lost. For digital how quality varies with is it is either on or off so signal strength is 0%, thus signal quality is 0% but when signal strength goes to 50% the signal quality is 100% as this is when it is on.

- Analogue can be converted into digital by 'sampling'.

Summary

Biology

Centres need to fully prepare learners for the exam by practicing exam technique, especially in relation to reading the question carefully and not repeating the stem of the question. Learners should also be taught that when they have answered the question to reread their response to ensure that the question set has been addressed in the answer they have given, and that they have used appropriate scientific knowledge and vocabulary.

Learners should have a clear understanding of the difference between “function” and “structure” and ensure they answer the question appropriately, for example if they are asked for the structure of an organelle, such as a plasmid, they don’t just provide the function.

Based on their performance on this paper, learners should:

- continue to practice exam technique, ensuring that they know what is required by specific command words and know the difference in what is required from commands such as describe and explain. Learners should expect to give a reason or linked expansion for explain questions, rather than just an identification point, as in 1(d) about the increase in speed in conduction caused by saltatory conduction.
- always show substitution into an equation then carry out any rearrangement and then the evaluation so that error carried forward marks can be awarded
- be familiar with and recognise the ultrastructure and function of organelles in eukaryotic cells (plant and animal cells) and eukaryotic cells (plant-cell specific), such as the cell wall and the vacuole
- learn about the structural differences between Gram-positive and Gram-negative bacterial cells and therefore why there is a difference in the action of antibiotics.
- use the information given in figures to support the response, such as in question 5.
- use the additional guidance to understand the depth and breadth of the specification

Chemistry

Based on their performance on this paper, learners should:

- ensure that they are aware of key terms such as property so that they can access questions appropriately
- ensure that they are familiar with the formula of common compounds and diatomic elements in the specification, that they know how to derive formula of unfamiliar compounds and that they balance equations

- practice exam techniques such as using the information in the stem of the question to inform and help them with their answer, perhaps underlining or highlighting key words to remind them as they are forming their answers
- ensure that they are accurately using subject specific terms and comparative language in their answers where necessary
- be taught that when a question asks them to explain they need to make their point and then given a justification of exemplification of the point, the answer must contain some reasoning
- ensure that when answers calculation questions that they show clear and logical working to help them arrive at the correct answer and so that error carried forward marks can be awarded if necessary.

Physics

Based on their performance on this paper, learners should:

- be able to convert units into SI format (for example, nm into m)
- practice laying out calculations and showing their working in a clear and logical way
- learn how the structure of an optical fibre allows it to perform its function
- practice applying knowledge to explain advantages and limitations of technology and signal types used in communications applications.

Please refer and link to the Applied Science Specification and/or Sample Assessment Materials (SAMs) on the BTEC Nationals Applied Science webpage, located [here](#).



Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

